AI IN HEALTHCARE · HIGH-RISK PROJECT · SP 2026

VLM-Powered Concept Bottleneck Models for Dermatology

with a Fitzpatrick Skin-Tone Fairness Audit

Deepesh Khanna

University of Texas at Austin

UID : dk32786

github.com/deepesh002/vlm-cbm-derm-fairness

A diagnosis that explains ***which clinical concept*** drove it *and where the disparity on darker skin really sits.*

Why this project and what we set out to answer

The problem

Skin-cancer classifiers underperform on darker skin *(Daneshjou et al., 2022)*

Grad-CAM shows ***where*** the model looked, not ***what*** it understood.

Fairness audits stop at accuracy and there is no handle on **which clinical feature** drives the gap.

Concept Bottleneck Models route every prediction through a handful of auditable concepts but have never been evaluated for skin-tone fairness.

Research questions

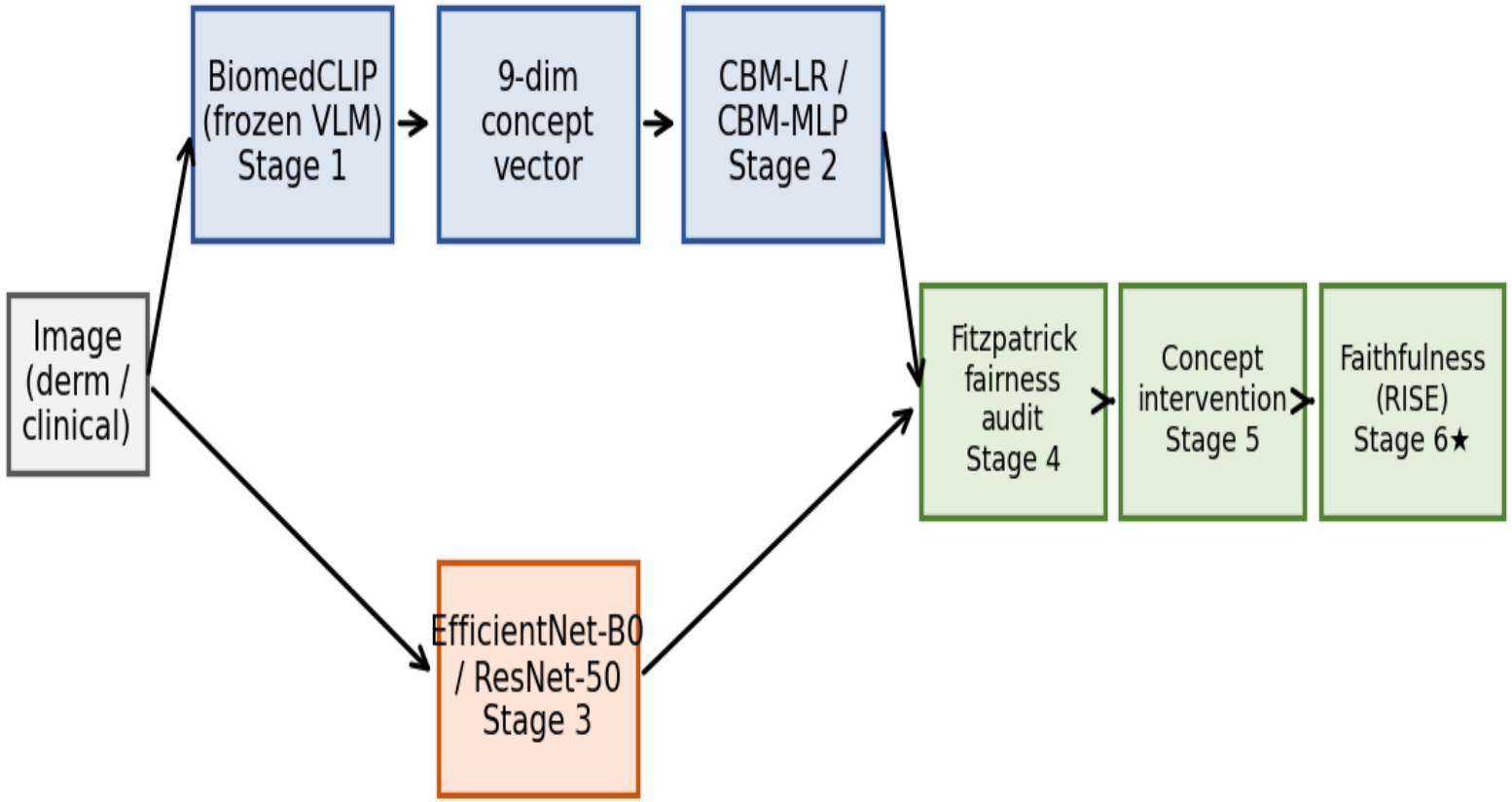
Q1 Can a concept-driven head match a deep baseline on melanoma classification?

Q2 Do errors concentrate in specific Fitzpatrick skin-tone buckets?

Q3 Do concept interventions rescue misdiagnoses and which concepts matter?

Q4 Are CBM explanations quantitatively more faithful than Grad-CAM?

End-to-end pipeline



1 **Concept prediction**

Frozen BiomedCLIP → 9 dermoscopic concepts via prompt pairs

2 **CBM head**

Sparse L_1 logistic regression on the concept probabilities

3 **Baselines**

EfficientNet-B0 & ResNet-50 on HAM10000 with Grad-CAM

4 **Fairness audit**

Per-concept & per-bucket AUROC, Spec, ECE, EOG, DPG

5 **Concept intervention**

Correct one concept at a time → measure rescue rate

6 **Faithfulness ★**

Insertion / deletion curves: concepts vs pixels

Four cohorts, four roles

Dataset	Size	Role in the audit	Fitzpatrick coverage
HAM10000	10,015	Primary training for EfficientNet-B0 & ResNet-50 baselines	Unlabeled (assumed light-skin skew)
Derm7pt	1,011 (294 pos)	CBM cross-validation + per-concept AUROC audit	No Fitzpatrick labels
DDI	656 (171 pos)	Fairness audit, expert-Fitzpatrick gold standard	Light I-II · Medium III-IV · Dark V-VI
Fitzpatrick17k	4,320 used	Pooled with DDI for the fairness audit (n = 4,976)	Self-reported + ITA-corrected Fitz I-VI

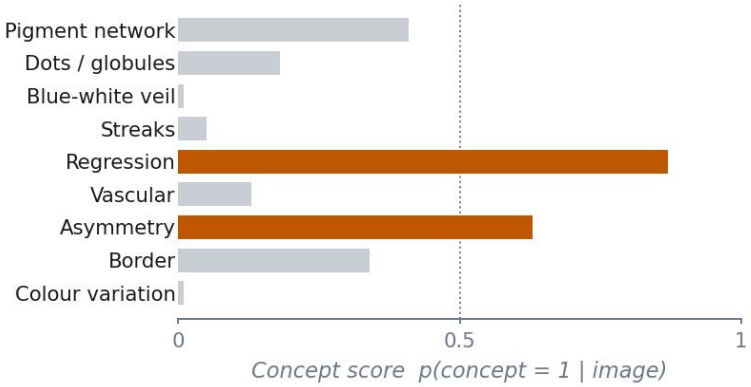
Why the pooled cohort matters. DDI alone gives a 100-sample dark-skin bucket; pooling Fitzpatrick17k lifts it to **n = 618** and stabilises every per-bucket metric. Bucket counts (light / medium / dark): **2,518 / 1,840 / 618.**

WORKED EXAMPLE

How the CBM reads a dermoscopy image

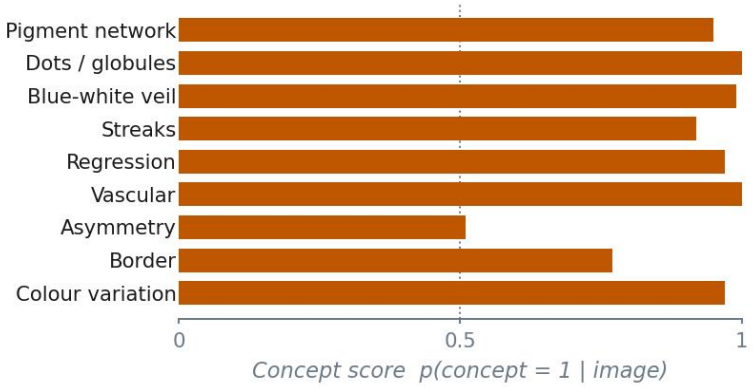
Benign lesion

Light skin (Fitzpatrick I-II)
CBM verdict: Benign



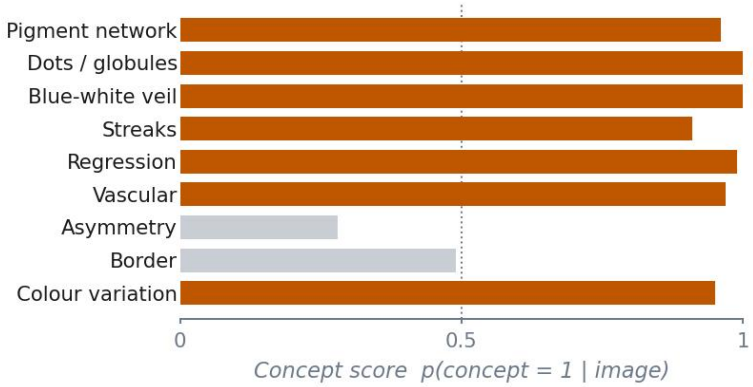
Melanoma

Light skin (Fitzpatrick I-II)
CBM verdict: Melanoma



Melanoma

Dark skin (Fitzpatrick V-VI)
CBM verdict: Melanoma



Read left-to-right. For every image, BiomedCLIP emits a probability per concept; the sparse head fires only when the right ones light up.

A 3-concept head matches a 9-concept model and a deep MLP

0.79

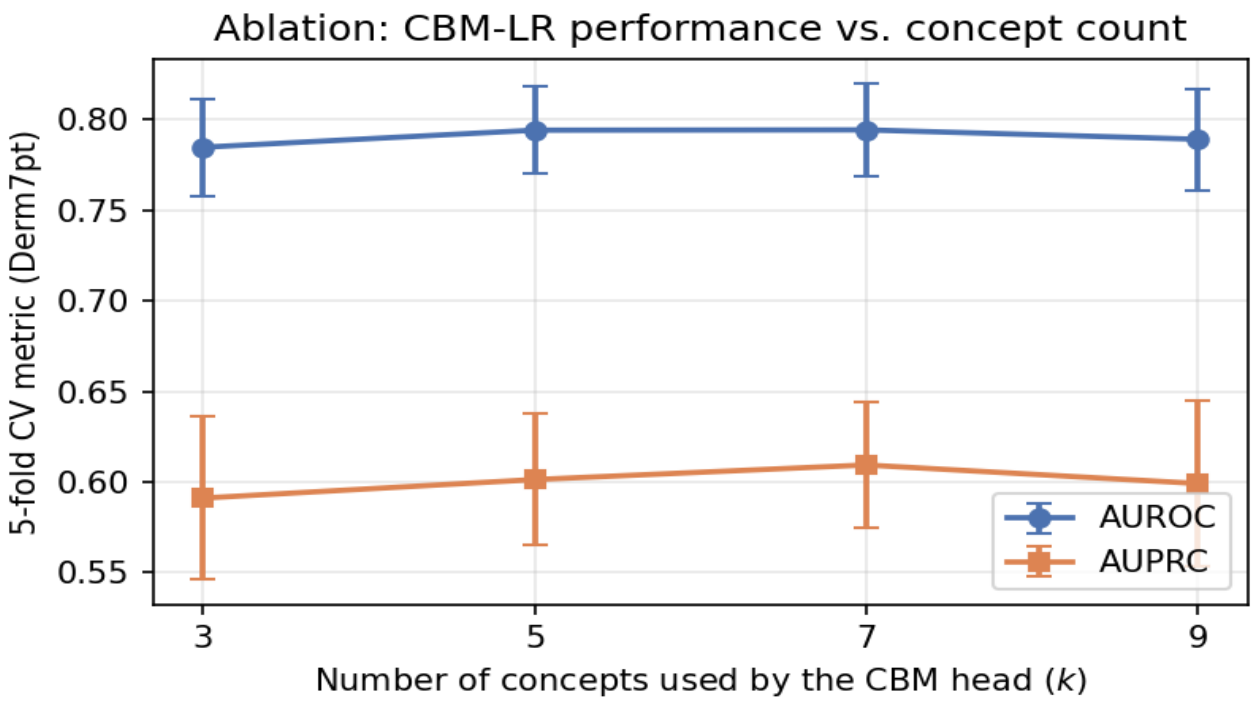
AUROC

5-fold CV on Derm7pt · ± 0.03

CBM-MLP head **0.79** *—> statistically indistinguishable*

Top-3 concepts only **0.785** *—> within 0.01 of the full 9*

Concept-count ablation

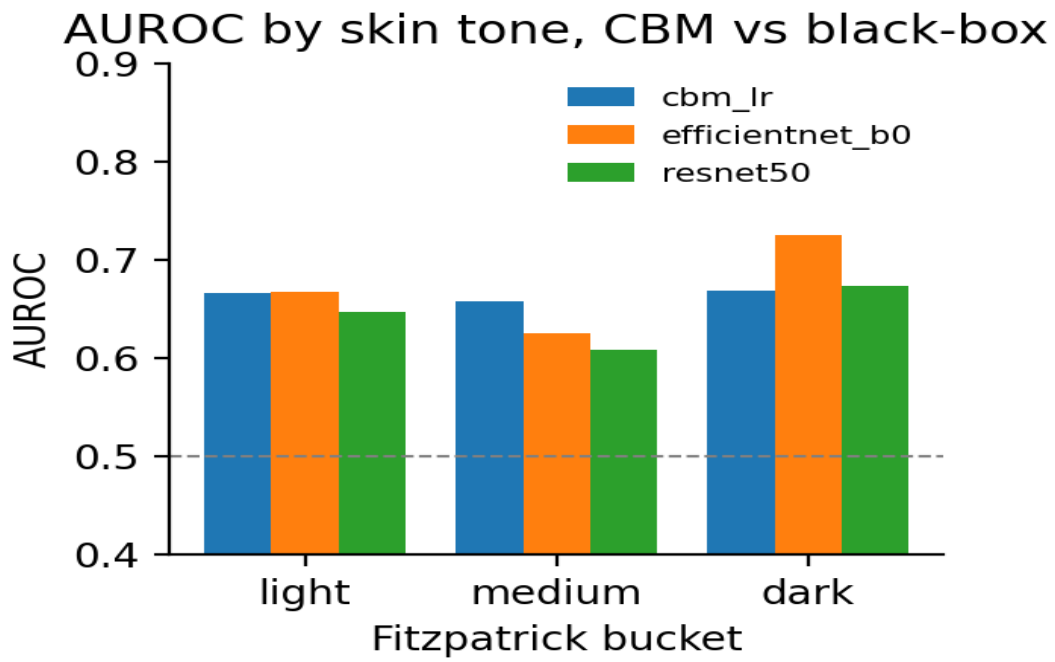


Takeaway. Three concepts saturate the head and this is the lever Q3 pulls on.

AUROC parity is not fairness

AT FIRST GLANCE

AUROC looks flat across skin tones



CBM-LR **0.67 / 0.66 / 0.67** *AUROC light / medium / dark*

BUT THEN

Specificity & calibration disagree

Bucket	AUROC	Spec.	ECE
Light I-II	0.67	0.58	0.080
Medium III-IV	0.66	0.56	0.103
Dark V-VI	0.67	0.45	0.178

Δ specificity light → dark: **−0.13**

Δ ECE light → dark: **+0.098**

Equalised-odds gap: **CBM 0.133** *> EffNet 0.098 > ResNet 0.051*

The head over-calls positives on dark skin rather than missing them and a clinically material fairness failure mode that AUROC alone hides.

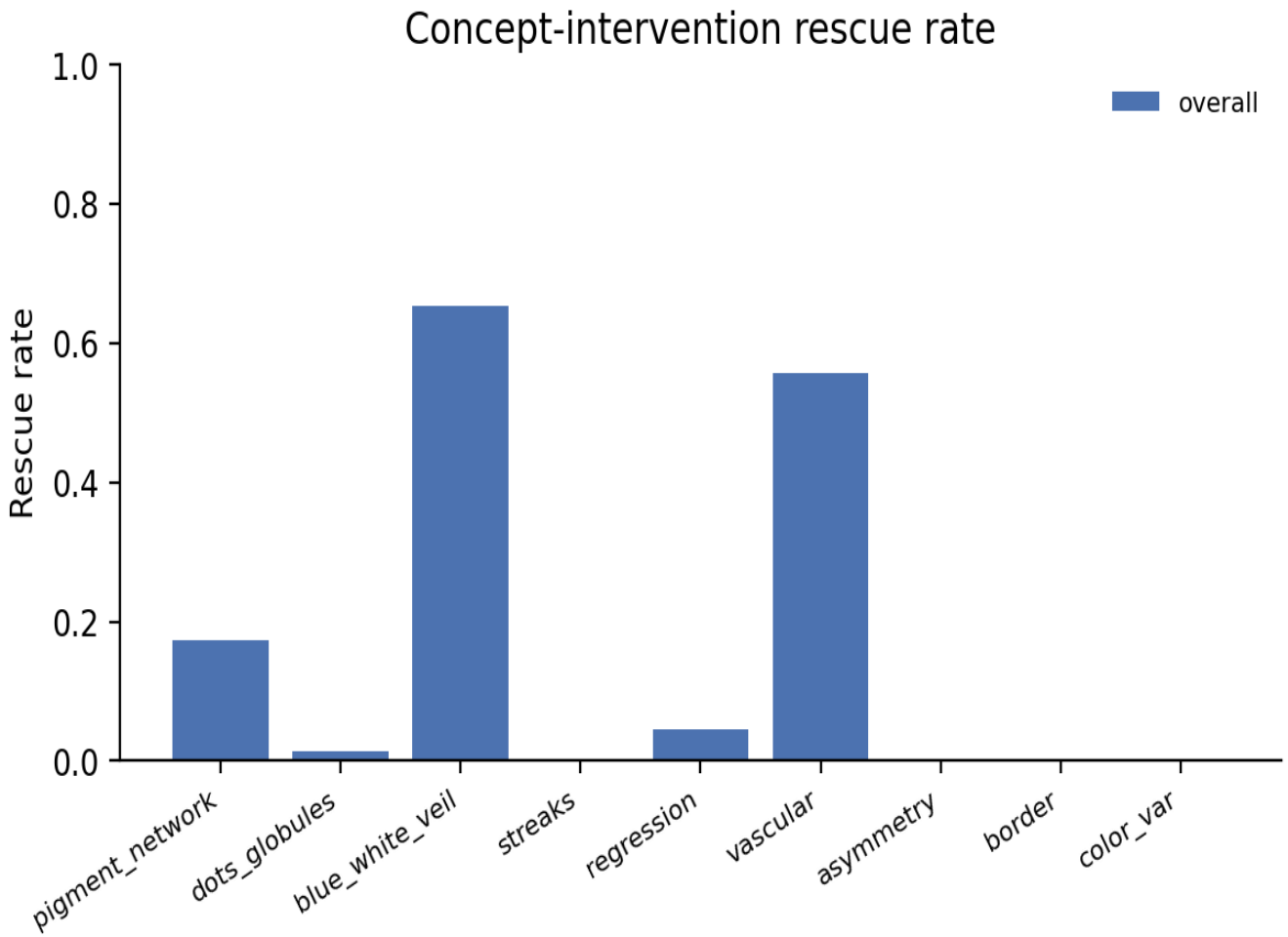
One concept fix rescues 85.8 % of CBM errors

85.8%

of CBM errors rescued by correcting ONE concept

86.5 % with two concepts corrected

Error budget lives in the concepts, not in the head.
Targeted concept retraining is the next highest-yield step.

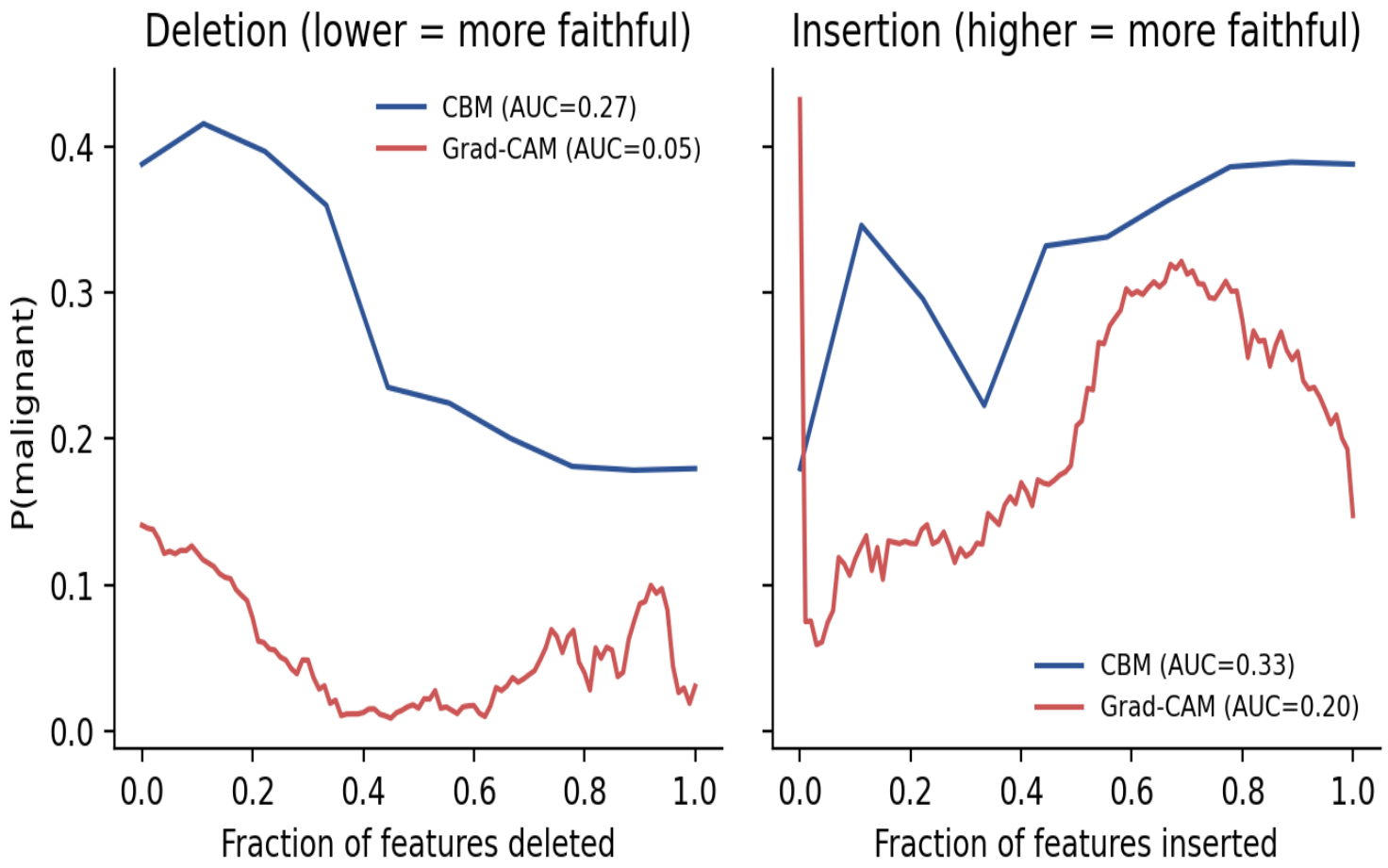


65.4 %
Blue-white veil

55.7 %
Vascular

17.3 %
Pigment network

Concepts beat Grad-CAM on insertion



CBM concepts

0.33

Insertion AUC (higher is more faithful)

dominant explanation

Grad-CAM

0.20

Insertion AUC (baseline saliency method)

insertion-AUC loss = 0.13

Deletion AUC is confounded by calibration differences and insertion is the decisive comparison.

Limitations and next steps

Limitations

VLM zero-shot concepts are noisy

Prompt-pair softmax is a weak concept classifier

Concept-layer audit still DDI-only

Fitz17k has no dermoscopic concept GT and the concept heatmap is unchanged

Fitz17k labels are self-reported

Self-report + ITA correction, not expert-annotated like DDI

Pooled bootstrap CIs not regenerated

Per-bucket CIs were DDI-only; pooled CIs are a clean next step

Future directions

1 **Targeted concept retraining**

Re-label blue-white-veil + pigment on Fitzpatrick V-VI imagery

2 **Per-bucket head recalibration**

Temperature / Platt scaling on dark-skin examples closes the ECE gap

3 **Learned linear probe on BiomedCLIP**

Quantify prompt-design vs. representation error

The CBM architecture turns a fairness gap into a concept-retraining task and not a dataset-collection moonshot.

CONTRIBUTIONS

Four takeaways from this audit

1

Interpretable accuracy

CBM-LR matches the MLP head (AUROC 0.79 on Derm7pt) and the top-3 concepts alone reach 0.785.

2

AUROC parity hides the real gap

On 4,976 pooled images AUROC is flat (0.67 / 0.66 / 0.67) but specificity drops 13 pts and ECE doubles on dark skin.

3

Concept intervention localises the error budget

85.8 % of errors fix on one concept; two-thirds via blue-white veil which is a directly actionable remediation target.

4

Faithful-by-construction explanations

Insertion fidelity 0.33 vs Grad-CAM 0.20 shows that the concept layer carries decision signal saliency cannot surface.

Thank you